

Seminar 2. Descriptive Statistics

Case: a trade and investment mission to Belgium

The situation

You have joined the analytical unit of a trade representation office. Your country is preparing a trade and investment mission to Belgium, and manufacturing is the focus of the visit. The delegation will meet company executives, an employers' federation and officials from the regional investment agency.

Three weeks before departure the head of the delegation sends your unit a short message:

“I have one dataset of Belgian production units and four questions. Do not write me an essay. I need numbers I can quote in the room”.

The four questions are these.

1. **What does a typical Belgian manufacturing firm look like?** The delegation should not prepare for meetings with industrial giants if it will in fact be sitting across from mid-sized firms.
2. **How concentrated is the sector?** If a handful of firms produce most of the output, the mission programme should be built around them.
3. **What is the wage level?** The argument "labour in Belgium is expensive" will be raised at the table, and the delegation needs to know what it actually costs.
4. **Do higher wages go together with higher productivity?** This is the standard reply to the previous argument, and it has to be checked rather than assumed.

Descriptive statistics answer all four. Nothing in this seminar requires any technique beyond the commands from the demonstration notebook.

How to work

Create a new notebook and place `Labour.csv` next to it. Import pandas and numpy. Add `pd.set_option("display.float_format", "{:.2f}".format)` so that large numbers are not printed in scientific notation.

Every task asks for a number and for one or two sentences of interpretation. The sentences matter as much as the numbers: the delegation will read them, not your code.

The dataset

`Labour.csv`, 569 Belgian production units, cross-section for 1996.

Column	Meaning	Unit
capital	total fixed assets at the end of 1995	millions of euro
labour	number of workers	persons
output	value added	millions of euro
wage	wage costs per worker	thousands of euro

Source of the data: <https://rdr.io/cran/Ecdat/man/Labour.html>. Original reference: Verbeek M. (2004) *A Guide to Modern Econometrics*, John Wiley and Sons, chapter 4.

Part A. Preparing the data

Task 1. Open the file and check it

Load `Labour.csv` into a dataframe. Report the number of rows and columns, the data type of every column, and the number of missing values in each column. Look at the first ten rows.

Question: how many firms are in the sample, and does any column need cleaning before the analysis starts?

Task 2. Print the minimum and the maximum of all four columns.

Question: do any of these eight numbers look like a data entry error rather than a real firm? State which value surprised you and why you decided to keep it or to doubt it.

Part B. What a typical firm looks like

Task 3. Two answers to the same question

Compute the mean and the median of all four columns, and then the difference between them.

Question: the head of the delegation asks for the size of a typical firm in one number. Do you report the mean number of workers or the median? Justify the choice in one sentence.

Task 4. How wide is the spread

For the `labour` column compute the standard deviation, the first and third quartiles and the interquartile range. Then compute the quartiles of `output`.

Question: compare the standard deviation of `labour` with its mean. What does the result tell you about the sector before you have drawn a single chart?

Task 5. The shape of the four distributions

Compute the skewness of all four columns and rank them from the most asymmetric to the least.

Question: which column is closest to symmetric, and why is that the one variable the delegation can discuss using an average?

Part C. Concentration and outliers

Task 6. Apply the outlier rule

For the `output` column compute the interquartile range and the bounds $Q1 - 1.5 * IQR$ and $Q3 + 1.5 * IQR$. Count how many firms fall outside those bounds and express the count as a share of the sample.

Question: a colleague proposes to delete these firms so that the averages "look normal". Give one argument against doing so in this particular case.

Task 7. Measure the concentration directly

Take the 10 per cent of firms with the largest `output` and compute what share of the total output of the sample they produce. Use `nlargest` and `sum`.

Question: answer question 2 of the delegation in one sentence, quoting your number.